

FSOC Data Rate calculations.
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Digital Communication

Baud Rate

Bandwidth

Modulation

Spectral Efficiency

Capacity

$$C = B \cdot \log_2 \left(1 + \frac{S}{N} \right)$$

Bit Energy

$$\frac{E_b}{N_0} =$$

Signal-to-Noise Ratio

$$\text{SNR}_0 = \frac{i_s}{\sigma_N} = \sqrt{\frac{\eta P_S}{2h\nu B}} \rightarrow 10 \cdot \log(\text{SNR}_0) = 74.9 \text{ dB}$$

$$i_S = \frac{\eta e P_s}{h\nu} = \frac{0.85 \cdot e \cdot 10 \text{ mW}}{h \cdot 193.4 \text{ THz}} = 0.01063$$

$$\langle i_N^2 \rangle = 2eBi_S = 2 \cdot e \cdot 100 \cdot 10^9 \cdot 0.01063 = 3.406 \cdot 10^{-10}$$

Bit Error Rate

$$\text{4-QAM: } BER \approx \cdot Q \left(\sqrt{\frac{2E_b}{N_0}} \right)$$

$$\text{16-QAM: } BER \approx 3 \cdot Q \left(\sqrt{\frac{4E_b}{5N_0}} \right)$$

OOK: On-Off Keying

QPSK: Quadrature Phase Keying

QAM: Quadrature Amplitude Modulation

Data Rate

Photodetectors

Noise

APD

PIN

Laser transmitter

Amplitude modulation

Phase modulation

Polarization modulation

Laser beam propagation

$$\theta = \frac{\lambda}{\pi w_0}$$

$$w(z) \approx w_0 + z\theta$$

$$w_{\text{LT}}^2(L) = w_{\text{vac}}^2(L) + 2.42k^{7/6}L^{11/6}w_0^{-1/3}C_n^2$$

Free-space

$$\Theta_0 = 1 - \frac{L}{F_0}$$

$$\Lambda_0 = \frac{2L}{kW_0^2}$$

$$W = W_0\sqrt{\Theta_0^2 + \Lambda_0^2}$$

$$I(0,L) = I_0 \cdot \frac{W_0^2}{W^2} \left[\frac{\text{W}}{\text{m}^2} \right]$$

$$P_{R_x} = P_{T_x} \cdot \left(1 - e^{\frac{-2r^2}{w^2}} \right) [\text{W}]$$

$$I_{R_x} = 4 \frac{P_{R_x}}{\pi D_{R_x}^2} \left[\frac{\text{W}}{\text{m}^2} \right]$$

Fried parameter

$$r_0 = (1.46 C_n^2 k^2 R)^{-3/5} \text{ for Horizontal link}$$

$$r_0 = \left[0.423 k^2 \sec(\theta) \int_{h_0}^H C_n^2(h) dh \right]^{-3/5} \text{ for Vertical link}$$

Refractive index structure parameter

HV: Hufnagel-Valley

$$C_n^2(h) = 0.00594(w/27)^2 * (10^{-5}h)^{10} \exp(-h/1000) + 2.7 \times 10^{-16} \exp(-h/1500) + A \exp(-h/100)$$

Greenwood frequency

$$f_g = 2.31 \lambda^{-6/5} \left[\sec(\theta) \int_{h_0}^H C_n^2(h) V_{\text{wind}}^{5/3}(h) dh \right]^{3/5}$$

Mean SNR

$$\langle \text{SNR} \rangle = \frac{\text{SNR}_0}{\sqrt{\left(\frac{P_{s0}}{\langle P_s \rangle} \right) + \sigma_I^2(D_G) \text{SNR}_0^2}}$$

$$\langle \text{SNR} \rangle \cong \frac{1}{\sigma_1(D_G)}, \quad \text{SNR}_0 \rightarrow \infty$$

Fading statistics

$$Pr_{\text{fade}} = 1 - \frac{1}{2} \int_0^\infty (u) \text{erfc} \left(\frac{\text{TNR} - \langle \text{SNR} \rangle u}{\sqrt{2}} \right) du$$

$$Pr(E) = \langle \text{BER} \rangle = \frac{1}{2} \int_0^\infty p_f(u) \text{erfc} \left(\frac{\langle \text{SNR} \rangle u}{2\sqrt{2}} \right) du$$

1 References

?? <https://www.nsnam.org/reviews/2016/socis-midterm/fso.html>